8 Ring Homomorphisms and Ideals

Definition 8.1.
Let R and S be rings. A map $\phi : R \to S$ is called a ring homomorphism if it satisfies

1. $\phi(a+b) = \phi(a) + \phi(b) \qquad \forall a, b \in R,$
   i.e. ϕ is a group homomorphism $\phi : (R, +) \to (S, +)$.
2. $\phi(ab) = \phi(a)\phi(b) \qquad \forall a, b \in R.$
3. $\phi(1_R) = 1_S$ if both R and S have identity.
   In particular $\phi(0_R) = 0_S$ and $\phi(-a) = -\phi(a)$ for all $a \in A$ since ϕ is a group homomorphism for $+$.
   A ring homomorphism which is a bijection is called an *isomorphism*.

Example 8.1.

1. The map
$$\phi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$$
$$a \ \mapsto \ \phi(a) = \bar{a}$$
is a ring homomorphism: $\phi(a+b) = \overline{a+b} = \bar{a} + \bar{b} = \phi(a) + \phi(b)$, $\phi(ab) = \overline{ab} = \bar{a}\bar{b} = \phi(a)\phi(b)$, and $\phi(1) = \bar{1}$.
2. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(a) = 2a$ is NOT a ring homomorphism: $\phi(ab) = 2ab \neq \phi(a)\phi(b) = 2a2b = 4ab$ if a and b are both non-zero.
3. The map
$$\phi : \mathbb{Q}[X] \to \mathbb{Q}$$
$$f(X) = a_0 + a_1X + \ldots + a_nX^n \ \mapsto \ \phi(f(X)) = f(0) = a_0$$
is a ring homomorphism (check this fact).
4. The map
$$\phi : \mathbb{Q}[X] \to \mathbb{Q}(\sqrt{2})$$
$$f(X) = a_0 + a_1X + \ldots + a_nX^n \ \mapsto \ \phi(f(X)) = f(\sqrt{2}) = a_0 + a_1\sqrt{2} + \ldots + a_n(\sqrt{2})^n$$
is a ring homomorphism.

Definition 8.2.
Let $\phi : R \to S$ be a ring homomorphism. We define
$$\mathrm{Ker}(\phi) = \{r \in R : \phi(r) = 0_S\}$$
called the *kernel* of ϕ, and
$$\mathrm{Im}(\phi) = \{s \in S : \exists r \in R \text{ with } \phi(r) = s\}$$
called the image of ϕ.

Example 8.2.

1. Consider the ring homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ defined by $\phi(a) = \bar{a}$. Then $a \in \mathrm{Ker}(\phi) \iff \phi(a) = \bar{a} = \bar{0} \iff n$ divides a, thus $\mathrm{Ker}(\phi) = n\mathbb{Z}$. Further $\mathrm{Im}(\phi) = \mathbb{Z}/n\mathbb{Z}$.

2. Consider the homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}$ defined by $\phi(f(X)) = f(0)$, i.e. if $f(X) = a_0 + a_1X + \ldots + a_nX^n$ then $\phi(f(X)) = a_0$ (see Examples 8.1).
Then
$$f(X) \in \mathrm{Ker}(\phi) \iff f(0) = 0 \iff a_0 = 0.$$
Thus
$$\mathrm{Ker}(\phi) = \{f(X) = a_1X + a_2X + \ldots + a_nX^n : a_i \in \mathbb{Q}, n \geq 1\}.$$
Further $\mathrm{Im}(\phi) = \mathbb{Q}$ since $a = \phi(a + X)$ for all $a \in \mathbb{Q}$.

Lemma 8.1.
Let $\phi : R \to S$ be a ring homomorphism. Then

1. $\mathrm{Ker}(\phi)$ *is a subring of* R.
2. $Im(\phi)$ *is a subring of* S.
3. ϕ *is surjective* $\iff Im(\phi) = S$.
4. ϕ *is injective* $\iff \mathrm{Ker}(\phi) = \{0_R\}$.

Proof.

1. Let $a, b \in \mathrm{Ker}(\phi)$. Then
$$\phi(a + b) = \phi(a) + \phi(b) = 0_S + 0_S = 0_S$$
and
$$\phi(ab) = \phi(a)\phi(b) = 0_S \times 0_S = 0_S,$$
hence $a + b, ab \in \mathrm{Ker}(\phi)$.
2. If $s_1 = \phi(r_1), s_2 = \phi(r_2)$ then
$$\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2) = s_1 + s_2$$
and
$$\phi(r_1r_2) = \phi(r_1)\phi(r_2) = s_1s_2,$$
hence $s_1 + s_2, s_1s_2 \in \mathrm{Im}(\phi)$.
3. Is clear.
4. Assume $\mathrm{Ker}(\phi) = \{0_R\}$. Let $r_1, r_2 \in R$ with $\phi(r_1) = \phi(r_2) \implies \phi(r_1) + (-\phi(r_2)) = \phi(r_1) + \phi(-r_2) = \phi(r_1 + (-r_2)) = 0_S \implies 0_R = r_1 + (-r_2) \in \mathrm{Ker}(\phi) \implies 0_R + r_2 = r_2 = r_1 + (-r_2) + r_2 = r_1 + ((-r_2) + r_2) = r_1 + 0_R = r_1$, hence $r_1 = r_2$.
Conversely suppose ϕ is injective. Let $r \in \ker(\phi)$. Then $\phi(r) = \phi(0_R) = 0_S$ hence $r = 0_R$ since ϕ is injective and $\mathrm{Ker}(\phi) = \{0_R\}$.

□

Definition 8.3.
Let R be a ring. A subset $I \subset R$ is called an *ideal* if the followings hold:

1. $(I, +)$ is a subgroup of $(R, +)$.
2. I is closed under multiplication by arbitrary elements of R:
$$ab \in I \qquad \forall a \in I \qquad \forall b \in R.$$

Example 8.3.

1. Let $m \in \mathbb{Z}$ then $m\mathbb{Z} = \{ma : a \in \mathbb{Z}\}$ is an ideal of $\mathbb{Z}$: if $k \in \mathbb{Z}, ma \in m\mathbb{Z}$ then $k(ma) = m(ka) \in m\mathbb{Z}$.
2. Let R be a ring and $a \in R$. The subset

$$(a)_R = \{ab : b \in R\} \subset R$$

is an ideal of R, called the *ideal generated by* a. Indeed, $0_R = a0_R \in (a)_R$, $ab + ac = a(b + c) \in (a)_R$, and $-(ab) = a(-b) \in (a)_R$. Further if $d \in R$ is arbitrary and $ac \in (a)_R$ then $d(ac) = dac = adc = a(dc) \in (a)_R$.

Lemma 8.2.
The kernel $\mathrm{Ker}(\phi)$ *of a ring homomorphism* $\phi : R \to S$ *is an ideal of* R.

Proof. We already know that $\mathrm{Ker}\,\phi$ is a subgroup of $(R, +)$. Let $a \in \mathrm{Ker}\,\phi$ and $b \in R$. Then $\phi(ab) = \phi(a)\phi(b) = 0_S\phi(b) = 0_S$ hence $ab \in \mathrm{Ker}(\phi)$. □

- Let $\phi : R \to S$ be a ring homomorphism and I and ideal of S then the inverse image

$$\phi^{-1}(I) = \{a \in R : \phi(a) \in I\}$$

is an ideal of R. Indeed, $\phi(0_R) = 0_S \in I \implies 0_R \in \phi^{-1}(I)$. If $c = \phi(a) \in I, d = \phi(b) \in I$ then $c + d = \phi(a) + \phi(b) = \phi(a + b) \in I \implies a + b \in \phi^{-1}(I)$. If $c = \phi(a) \in I$ then $-c = \phi(-a) \in I \implies -a \in \phi^{-1}(I)$ since $-c \in I$. Finally let $a \in \phi^{-1}(I)$ and $b \in R$. Then $\phi(a) \in I$ and $\phi(ab) = \phi(a)\phi(b) \in I$ since I is an ideal.

Definition (Construction of the quotient ring R/I).

Let R be a ring and I an ideal of R. Recall the relation $\mathcal{H}$ among elements of R defined by (see Definition 2.1):

$$a \mathcal{H} b \iff b - a \in I.$$

The relation $\mathcal{H}$ is an equivalence relation (see Lemma 2.1). The equivalence class of $a \in R$ for the relation $\mathcal{H}$ is $\bar{a} = \{b \in R : b = a + c, c \in I\}$. Thus

$$\bar{a} = a + I = \{a + c : c \in I\}$$

is called the *class of* a *modulo* I [this is also the left coset $a + I$ of the subgroup $(I, +)$ of $(R, +)$]. Set

$$R/I = \{\bar{a} : a \in A\} = \{a + I : a \in R\}$$

for the set of equivalence classes of elements of R modulo I, this is the set of left cosets of the subgroup $(I, +)$ in R. Note that

$$\bar{a} = \bar{b} \iff a - b \in I.$$

Any element in $\bar{a}$ is called a *representative* of the class of a modulo I.

We define addition and multiplication among elements of R/I by the formulas

$$\bar{a} + \bar{b} = \overline{a + b} \qquad (a + I) + (b + I) = a + b + I,$$

and

$$\bar{a} \times \bar{b} = \overline{ab} \qquad (a+I)(b+I) = ab + I.$$

(Check these are well defined and independent of the choice of a representative of a given class, for $+$ see Definition 2.5.)

$(R/I, +, \times)$ is a ring (check this fact) called the *quotient ring* of R by the ideal I. The neutral element for addition is

$$0_{R/I} = \overline{0_R} = 0_R + I = I.$$

If R has an identity 1_R then R/I has an identity as well and

$$1_{R/I} = \overline{1_R} = 1_R + I$$

since $\overline{1_R} \times \bar{a} = \bar{a}$ for all $a \in A$.

Lemma 8.3.
Consider the map $\phi : R \to R/I$ defined by $\phi(a) = \bar{a}$ for $a \in R$. Then

1. *ϕ is a surjective ring homomorphism.*
2. $\mathrm{Ker}(\phi) = I$.

Proof. 1) follows immediately from the definition of ϕ (see also Proposition 2.7 where we proved ϕ is a homomorphism for $+$). For 2): $a \in \mathrm{Ker}(\phi) \iff \bar{a} = \overline{0_R} \iff a - 0_R = a \in I$ (see Proposition 2.7). □

Example 8.4.

1. Let $n \geq 1$ an integer and $I = n\mathbb{Z} = \{na : a \in \mathbb{Z}\}$ the principal ideal of $\mathbb{Z}$ generated by n. The relation $\mathcal{H}$ in this case is the relation of congruence modulo n and $\mathbb{Z}/I = \mathbb{Z}/n\mathbb{Z}$ endowed with $+$ and $\times$: $\bar{i} + \bar{j} = \overline{i+j}$, $\bar{i} \times \bar{j} = \overline{ij}$, is a ring.
2. Let $R = \mathbb{Z}[X]$ and consider the principal ideal of R generated by X^2:

$$I = (X^2)_R = \{h(X) = X^2(b_2 + b_3X + \ldots b_nX^{n-2}) \ : \ b_i \in \mathbb{Z}, n \geq 2\}.$$

$$I = \{h(X) = b_2X^2 + b_3X^3 + \ldots b_nX^n \ : \ b_i \in \mathbb{Z}, n \geq 2\}$$

Let $f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_mX^m \in \mathbb{Z}[X]$. Then $f(X) - (a_0 + a_1X) = a_2X^2 + \ldots + a_mX^m \in I$ which implies $\overline{f(X)} = \overline{a_0 + a_1X}$. Hence every class of a polynomial in $\mathbb{Z}[X]$ modulo I is represented by the class of a polynomial of degree ≤ 1 and

$$\mathbb{Z}[X]/I = \{\overline{a_0 + a_1X} : a_0, a_1 \in \mathbb{Z}\}.$$

In $\mathbb{Z}[X]/I$ we have

$$\overline{3 + 5X + X^4} = \overline{3 + 5X + X^3 - X^5},$$
$$(\overline{1 + 3X}) + (\overline{-4 + 5X}) = \overline{-3 + 8X},$$

and

$$(\overline{1 + 3X})(\overline{-4 + 5X}) = \overline{-4 - 7X + 15X^2} = \overline{-4 - 7X}.$$

3. Let $R = \mathbb{Q}[X]$ and

$$I = (X^2 - 2)_{\mathbb{Q}[X]} = \{(X^2 - 2)f(X) : f(X) \in \mathbb{Q}[X]\}.$$

Let $g(X) \in \mathbb{Q}[X]$. Then $g(X) = (X^2 - 2)f(X) + (a_0 + a_1X)$ (Euclidean division), and $g(X) - (a_0 + a_1X) \in I \iff \overline{g(X)} = \overline{a_0 + a_1X}$ where $a_0 + a_1X$ is the rest of the division of $g(X)$ by $X^2 - 2$. Hence

$$\mathbb{Q}[X]/I = \{\overline{a_0 + a_1X} : a_0, a_1 \in \mathbb{Q}\}.$$

Theorem 8.4 (First Isomorphism Theorem for Rings).
Let $f : R \to S$ be a surjective ring homomorphism. Then $R/\operatorname{Ker}(f)$ is isomorphic to S.

Proof. We consider the map $\pi : R/\operatorname{Ker}(f) \to S$ defined by $\pi(\bar{a}) = f(a)$ for $a \in R$. First we check π is well defined. Assume $\bar{a} = \bar{b} \iff a-b \in \operatorname{Ker}(f) \iff f(a-b) = 0_S \iff f(a) = f(b) \iff \pi(\bar{a}) = \pi(\bar{b})$.

Next we show that π is a homomorphism: $\pi(\bar{a}+\bar{b}) = \pi(\overline{a+b}) = f(a+b) = f(a)+f(b) = \pi(\bar{a})+\pi(\bar{b})$ and $\pi(\bar{a}\bar{b}) = \pi(\overline{ab}) = f(ab) = f(a)f(b) = \pi(\bar{a})\pi(\bar{b})$.

We check π is bijective. First π is surjective since given $s \in S$ there exists $a \in R$ with $s = f(a)$ (as f is surjective) hence $s = \pi(\bar{a})$. To show π is injective we show $\operatorname{Ker}(\pi) = \{\bar{0}_R\}$. Let $a \in R$ with $\pi(\bar{a}) = f(a) = 0_S$. Then $a \in \operatorname{Ker}(f)$ which implies $\bar{a} = \bar{0}_R$.

Alternatively, using the First Isomorphism Theorem for groups, see 2.10, we constructed $\pi : R/\operatorname{Ker}(f) \to S$ and showed it is an isomorphism of groups for $+$. Thus we just need to show π is a ring homomorphism by showing $\pi(\bar{a}\bar{b}) = \pi(\bar{a})\pi(\bar{b})$ which we did above. □

Example 8.5.

1. The homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}$ defined by
$$\phi(f(X)) = f(0)$$
is surjective (see Examples 8.1). We compute $\operatorname{Ker}(\phi)$. We have $\operatorname{Ker}(\phi) = \{f(X) = a_0 + a_1X + \ldots + a_nX^n \in \mathbb{Q}[X] : f(0) = a_0 = 0\} = \{f(X) = a_1X + \ldots + a_nX^n \in \mathbb{Q}[X]\} = \{X(a_1 + a_2X + \ldots + a_nX^{n-1}) \in \mathbb{Q}[X]\} = (X)_{\mathbb{Q}[X]}$ is the principal ideal of $\mathbb{Q}[X]$ generated by X. The first isomorphism theorem for rings implies that $\mathbb{Q}[X]/(X)$ is isomorphic to $\mathbb{Q}$.
2. The homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}(\sqrt{2})$ defined by
$$\phi(f(X)) = f(\sqrt{2})$$
[if $f(X) = a_0 + a_1X + \ldots + a_nX^n$ then $f(\sqrt{2}) = a_0 + a_1\sqrt{2} + \ldots + a_n(\sqrt{2})^n$] is a surjective homomorphism. Indeed if $x = a + b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ then $x = \phi(a + bX)$.
We claim $\operatorname{Ker}(\phi) = I = (X^2 - 2)_{\mathbb{Q}[X]}$ is the principal ideal I generated by $X^2 - 2$. Let $f(X) = (X^2-2)g(X) \in I = (X^2-2)_{\mathbb{Q}[X]}$, then $\phi(f) = \phi(X^2-2)\phi(g) = (\sqrt{2}^2-2)g(\sqrt{2}) = 0$ and $f \in \operatorname{Ker}(\phi)$. Conversely let $f \in \operatorname{Ker}(\phi)$. Then $f(X) = (X^2-2)g(X)+r(X)$ with $r(X) = a+bX$ [$\deg r < \deg(X^2 - 2) = 2$]. In particular, $0 = f(\sqrt{2}) = (\sqrt{2}^2 - 2)g(\sqrt{2}) + a + b\sqrt{2} = 0 + a + b\sqrt{2} \implies a = -b\sqrt{2} \implies a = b = 0$ and f is divisible by $X^2 - 2$ hence $f \in I$.
The first isomorphism theorem for rings implies that $\mathbb{Q}[X]/(X^2 - 2)$ is isomorphic to $\mathbb{Q}(\sqrt{2})$

Definition 8.4.
Let R be a ring and I, J, ideals of R. Set
$$I + J = \{a + b : a \in I, b \in J\}$$
and
$$IJ = \{\sum_{i=1}^{m} a_ib_i : a_i \in I, b_i \in J, m \geq 1\}.$$

Lemma 8.5.
Let R be a ring and I, J ideals of R. Then $I + J$ and IJ are ideals of R.

Proof. will do in lecture. □

Definition 8.5.
Let R be a ring.

1. An ideal I of R is called *principal* if $I = (a)_R$ is generated by one element $a \in R$, called a *generator* of I. Recall $(a)_R = \{ab : b \in R\}$.
2. Let $a, b \in R$ and define $(a, b)_R = \{ac + bd : c, d \in R\} = (a)_R + (b)_R$. Then $(a, b)_R$ is an ideal (being the sum of two ideals) called the ideal *generated* by a and b.

Lemma 8.6.
Let R be a ring.

1. *$(b)_R \subset (a)_R \iff b = ac$ for some $c \in R$.*
2. *If R is an integral domain with identity, then*

$$(a)_R = (b)_R \iff a = bu : u \in R^\times.$$

Proof. Will do in lecture. □

Example 8.6 (Example of a non principal ideal)**.**
Let

$$R = \mathbb{Z}[X]$$

and

$$I = (2, X)_{\mathbb{Z}[X]} = \{2h(X) + Xg(X), h, g \in \mathbb{Z}[X]\}$$

the ideal of $\mathbb{Z}[X]$ generated by 2 and X. I claim I is NOT principal.

We argue by contradiction. Indeed assume the contrary

$$I = (f(X))_{\mathbb{Z}[X]}$$

is principal generated by some $f(X) \in \mathbb{Z}[X]$.

Then

$$2 = 2 \times 1 + X \times 0 \ \in \ I$$

hence $\exists p(X) \in R$ such that

$$2 = f(X)p(X).$$

In particular $0 = \deg 2 = \deg(f) + \deg(p)$ which implies $\deg(f) = \deg(p) = 0$ and both f and p are constant polynomials.

Further since f divides 2 it holds that $f \in \{1, -1, 2, -2\}$. If $f = 1$ or $f = -1$ then

$$1 = 2h(X) + Xg(X)$$

for some polynomials $h, g \in \mathbb{Z}[X]$ which is not possible (the constant coefficient of $2h(X) + Xg(X)$ is even). So $f = 2$ or $f = -2$. But

$$X = 0 \times 2 + 1 \times X \ \in \ I$$

hence $X = 2q(X)$ for some $q(X) \in \mathbb{Z}[X]$, which again is not possible.

Example 8.7 (Construction of a surjective homomorphism with a given kernel)**.**
Let

$$R = \mathbb{Z}[\sqrt{-3}] = \{a + b\sqrt{-3} : a, b \in \mathbb{Z}\}$$

and

$$J = (2, 1 + \sqrt{-3})_R = \{2x + (1 + \sqrt{-3})y : x, y \in \mathbb{Z}[\sqrt{-3}]\}.$$

Construct a surjective homomorphism from R to a finite field with kernel J.

Consider the map

$$f : \mathbb{Z}[\sqrt{-3}] \ \to \ \mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$$

defined by

$$a + b\sqrt{-3} \quad \mapsto \quad f(a + b\sqrt{-3}) = \bar{a} + \bar{b}.$$

This map is surjective as $\bar{1} = f(1)$ and $\bar{0} = f(2)$.

- We show f is a ring homomorphism.
  Indeed
  $f(a + b\sqrt{-3} + c + d\sqrt{-3}) = f(a + c + (b + d)\sqrt{-3}) = \overline{a + c} + \overline{b + d} = \bar{a} + \bar{b} + \bar{c} + \bar{d} = f(a + b\sqrt{-3}) + f(c + d\sqrt{-3})$
  and
  $f((a+b\sqrt{-3})(c+d\sqrt{-3})) = f(ac-3bd+(ad+bc)\sqrt{-3}) = \overline{ac - 3bd}+\overline{ad + bc} = \overline{ac + bd}+\overline{ad + bc} = \overline{ac} + \overline{bd} + \overline{ad} + \overline{bc} = \bar{a}\bar{c} + \bar{b}\bar{d} + \bar{a}\bar{d} + \bar{b}\bar{c} = (\bar{a} + \bar{b})(\bar{c} + \bar{d}) = f(a + b\sqrt{-3})f(c + d\sqrt{-3})$ (here we used for the third equality the fact that $\overline{-3} = \bar{1}$ in $\mathbb{F}_2$).
- We show $\mathrm{Ker}(f) = (2, 1 + \sqrt{-3})_R$.
  Clearly $J = (2, 1 + \sqrt{-3})_R \subset \mathrm{Ker}(f)$ since $f(2) = f(1 + \sqrt{-3}) = \bar{0}$ hence $f(2x + (1 + \sqrt{-3})y) = f(2)f(x) + f(1 + \sqrt{-3})f(y) = 0f(x) + 0f(y) = 0$ for $x, y \in R$.
  We show that $\mathrm{Ker}(f) \subset (2, 1 + \sqrt{-3})_R$. Let $a + b\sqrt{-3} \in \mathrm{Ker}(f)$ so that $f(a + b\sqrt{-3}) = \bar{a} + \bar{b} = \overline{a - b} = \bar{0}$. Hence $a - b = 2k$ and $a = b + 2k$ for some $k \in \mathbb{Z}$. Thus $a + b\sqrt{-3} = 2k + b + b\sqrt{-3} = 2k + b(1 + \sqrt{-3}) \in J$ as required.